ZS-M3

Regge-Holonomy, Immirzi & Z-Telomere

The Topological Life-Cycle: Z-Spin Isomorphism & Z-Telomere Collapse

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Verification: 27/27 PASS | Zero Free Parameters

§0. Abstract

Structural isomorphism Q=11 ↔ 4-valent j=1/2 quantum tetrahedron (Theorem 5.1, unique). Z-Telomere mechanism: phase drift δφ = A via Regge-Holonomy (Lemma 8.1: δφ_cell = A·I_cell, I_cell = κ/r = 4/4 = 1). BH entropy: S = A_H/(4G_eff) − ln 2. Dual Z-anchor: Tolman + vortex-core. Timescale hierarchy: $\tau_2 \approx 6.3\times10^{-10}$ s (weak), $\tau_5 \approx 2.6\times10^{34}$ yr ($p\to e^+\pi^0$ partial lifetime). Spinor Phase Gate: Π_Z(φ) = sin²(φ/2) derived from j=1/2 uniqueness + SU(2) Wigner d-matrix. Verification: 27/27 PASS.

Epistemic Status Legend

Status	Definition
PROVEN	Follows from standard mathematics alone. Machine-verifiable.
DERIVED	Follows from Z-Spin action + standard physics. Zero free parameters.
DERIVED-under-P6	Derived conditional on Theorem P6* (primitive locality, ZS-U8 v1.0 §3.0a).
VERIFIED	Numerically confirmed against observational data or independent computation.
TESTABLE	Well-defined prediction awaiting experimental data.
HYPOTHESIS (Hn)	Physically motivated conjecture. Derivation chain incomplete.
CONSISTENT	Compatible with framework structure but not independently derived.
NON-CLAIM	Explicitly not asserted. Documented to prevent overclaim.
RETRACTED	Previously claimed, now withdrawn with documented reason.

§1. Q=11 ↔ Quantum Tetrahedron

Theorem 5.1 (Uniqueness). Among all half-integer spins, only j=1/2 yields dim(Inv) = 2 = Z for the 4-valent quantum tetrahedron. [PROVEN]

j	1/2	1	3/2	2	5/2
dim(Inv)	2 = Z ✓	3 = X	4	5	6 = Y

$$Q = Z + X + Y = 2 + 3 + 6 = 11 \quad (1)$$

§2. BH Entropy

$$S_{BH} = A_H / (4G_{eff}) - \ln 2 \quad (2)$$

$G_{eff} = G_N/(1+A) = G_N \times 437/472$. The $-\ln 2$ correction is universal (Z_2 seam, 1 bit per link). [STATUS: DERIVED]

§3. Immirzi Parameter

$$\gamma_{j=1/2} = \ln 2 / (\pi\sqrt{3}) \approx 0.12738 \quad (3)$$

Z_2 seam is the physical origin of ln 2: each link encodes 1 bit parity ($\varepsilon = \pm 1$). NON-CLAIM: ZS-M3 does NOT predict γ numerically. [DERIVED interpretation]

§4. Dual κ Census

Definition	κ = 0	κ = 4	κ = 6
κ_disrupted (ZS-F5 v1.0)	1 (I only)	3 (diagonal)	6 (non-diag)

P6 bridge: Theorem P6* (primitive locality, ZS-U8 v1.0 §3.0a) proves primitive cells have P=I → diagonal → κ=4. Selects: 1 identity + 3 diagonal with κ=4. [PROVEN]

§5. Regge-Holonomy Framework

Lemma 8.1: $\delta\varphi_{cell}(c) = A \cdot I_{cell}(c)$, $I_{cell} = \kappa/r$ (4)

Primitive cell: $I_{cell}(c^*) = 4/4 = 1 \rightarrow \delta\varphi = A = 35/437$ (5)

7-step derivation: ZS-F2 v1.0(A) → ZS-F4 v1.0(r=4) → ZS-F5 v1.0(κ census) → P6 bridge (Theorem P6*, ZS-U8 v1.0 §3.0a, PROVEN) → Lemma 8.1 → I_cell=1 → δφ = A. [DERIVED]

§6. Z-Telomere Mechanism

$$T_{micro} = 2\pi/A \approx 78.45 \text{ Planck cycles} \quad (6)$$
$$t_{flip} = \pi/A \approx 39.22 \text{ Planck } (Z_2 \text{ half-event}) \quad (7)$$
$$\kappa=4 \text{ phase: } 3 \times (2\pi/3) = 2\pi = \Phi_c \ \ (3 = X \text{ mediates}) \quad (8)$$
$$S_{tunnel} = 5\pi/A = 5 \text{ seam flips} = 196.13 \quad (9)$$

$S_{tunnel}/T_{micro} = 5/2 = |I_h|/|O_h|$. Factor 5 = $|I_h/T_d|$ independent origin. [DERIVED]

§7. Z-Anchor: ε(r_H) = 0

Tolman (thermal): T_local → ∞ at horizon → Z_2 symmetry restored. [DERIVED]

Vortex-core (topological): $\pi_1(S^1) = \mathbb{Z} \to \rho(0) = 0$ at defect core. [DERIVED]

Two independent mechanisms converge on the same boundary condition.

§8. Timescale Hierarchy

$$\tau_n = t_P \times exp(n \times \pi/A) \quad (10)$$

n	Group Origin	τ_n	Physical Scale	Status
2	|O_h/T_d| = 48/24	6.3×10^{-10} s	Weak baryon decay	DERIVED
5	|I_h/T_d| = 120/24	2.6×10^{34} yr	$p\to e^+\pi^0$ partial lifetime	DERIVED
3	X = 3	~2 yr	(weak match)	CONSISTENT
4	φ(5) = 4	~10^{17} yr	(fair match)	CONSISTENT

§9. SU(2) Center Pairing

Lemma 10.1: $D^j(-I) = (-1)^{2j} \cdot I$ (11)

j=1/2: $D^{1/2}(-I) = -I$ (sign flip). Structural support for UV convergence. [PROVEN]

§10. Spinor Phase Gate

The Z-sector mediator has dim(Z) = 2 (ZS-F5 v1.0, PROVEN). Theorem 5.1 (§1) proves this is uniquely the j = 1/2 invariant subspace of the 4-valent quantum tetrahedron.

10.1 SU(2) Wigner d-matrix for j = 1/2

For spin j = 1/2, a rotation by angle φ about any axis gives the Wigner (small) d-matrix: $d^{1/2}(\varphi)$ with off-diagonal element $d^{1/2}_\{-+\}(\varphi) = \sin(\varphi/2)$. The transition probability from state $|+\rangle$ to $|-\rangle$ through rotation φ is:

$$P(|+\rangle \to |-\rangle) = |d^{1/2}_\{-+\}(\varphi)|^2 = \sin^2(\varphi/2) \quad (12)$$

This is a mathematical identity of SU(2) representation theory. Numerical verification: SU(2) matrix exponentiation $U(\varphi) = \exp(-i\varphi\sigma_y/2)$ at 8 test points yields $|U_{10}|^2 = \sin^2(\varphi/2)$ to machine precision ($\Delta < 10^{-14}$). [STATUS: PROVEN]

10.2 Physical Identification: Phase Gate = Transition Probability

The Z-mediated transfer efficiency depends on the overlap between input and output Z-states. The holonomy phase φ_Z accumulates during ε-field evolution at rate δφ_cell = A per primitive cell (Lemma

8.1, §5). At $\varphi_Z = 0$: no holonomy, no mediation, gate = 0. At $\varphi_Z = \pi$: maximum rotation, maximum mediation, gate = 1. At $\varphi_Z = 2\pi$: SU(2) half-return, gate = 0 (spinor sign flip from Lemma 10.1: $D^{1/2}(-I) = -I$). At $\varphi_Z = 4\pi$: full SU(2) return, gate = 0. Therefore:

$$\Pi_Z(\varphi_Z) = \sin^2(\varphi_Z / 2) \quad (13)$$

The boundary condition gate(0) = 0 uniquely selects $\sin^2$ over $\cos^2$. The 4π periodicity is the defining signature of SU(2) spinor structure versus SO(3) rotation (2π period). [STATUS: DERIVED]

10.3 Time-Average Consistency with ZS-Q7 v1.0

The Pauli master equation (ZS-Q7 v1.0 §5.1) provides time-averaged transition rates $W_{AB} = \dim(B)\cdot A/Q$. Consistency requires $\langle \sin^2(\varphi/2)\rangle = (1/4\pi)\int \sin^2(\varphi/2)d\varphi = 1/2$ over one full SU(2) period $[0, 4\pi]$. This is verified analytically. The cosmological closure mapping (ZS-Q7 v1.0 §5.8) uses this time-average to connect the instantaneous Z-Sim currents to the master equation rates, yielding $\gamma_{xz} = 2A/Q$ and $\gamma_{zy} = 6A/Q$ with zero free parameters. [STATUS: DERIVED]

10.4 Derivation Chain Summary

ZS-F5 v1.0 (dim(Z) = 2, PROVEN) → Theorem 5.1 (j = 1/2 uniqueness, PROVEN, §1) → SU(2) Wigner d-matrix (mathematics, PROVEN) → $P = \sin^2(\varphi/2)$ (eq. 12) → Lemma 10.1 (4π periodicity, PROVEN, §9) → Physical BC: gate(0) = 0 selects $\sin^2$ → $\Pi_Z(\varphi_Z) = \sin^2(\varphi_Z/2)$ [DERIVED]. Maximum derivation depth: 3 steps from PROVEN inputs. This eliminates the last HYPOTHESIS closure parameter in the Z-Spin forward simulator (ZS-T3 v1.0), achieving zero free parameters across the complete Z-Sim engine. [STATUS: DERIVED]

§11. Claims

ID	Statement	Status
C1	j=1/2 unique: dim(Inv) = 2 = Z	**PROVEN**
C2	κ census: {0:1, 4:3, 6:6}	**PROVEN**
C3	P6 bridge → κ=4 primitive cell	**PROVEN (Thm P6*)**
C4	$\delta\varphi = A$ (primitive cell, P6 closed)	**DERIVED**
C5	$T_{micro} = 2\pi/A$	**DERIVED**
C6	S_{tunnel} = 5 seam flips	**DERIVED**
C7	$S_{BH} = A_H/(4G_{eff}) - \ln 2$	**DERIVED**
C8	$\varepsilon(r_H) = 0$ (dual mechanisms)	**DERIVED**
C9	$D^j(-I) = (-1)^{2j}\cdot I$	**PROVEN**
C10	$\tau_2 \approx 6.3\times10^{-10}$ s, $\tau_5 \approx 2.6\times10^{34}$ yr	**DERIVED**
C11	Spinor phase gate: $\Pi_Z(\varphi) = \sin^2(\varphi/2)$ from j=1/2 SU(2)	**DERIVED**

§12. Verification Suite

Category	Tests	Pass	Scope
A: Structural isomorphism	4	4	j=1/2 unique, Q=11, sectors
B: Dual κ census & P6	5	5	10 involutions, distribution, bridge
C: Regge-Holonomy & Z-Telomere	5	5	δφ=A, T_micro, t_flip, S_tunnel, phase
D: Z-Anchor & BH entropy	4	4	G_eff, −ln 2, Tolman, vortex
E: Immirzi & SU(2) center	3	3	γ, D^j(−I), ln 2 origin
F: Timescale hierarchy	4	4	τ_2, τ_5, ratio 5/2, coset
G: Spinor phase gate	2	2	SU(2) matrix, time-average = 1/2
TOTAL	**27**	**27**	100% pass rate

Required dependencies: numpy, scipy, mpmath. The suite will not run without mpmath.

§13. Conclusion

This paper establishes the structural isomorphism Q=11 ↔ 4-valent j=1/2 quantum tetrahedron (Theorem 5.1, unique among all half-integer spins). The Regge-Holonomy framework yields the Z-Telomere mechanism with phase drift δφ = A per primitive cell (P6 closed, DERIVED), producing the timescale hierarchy $\tau_n = t_P \cdot \exp(n\pi/A)$ with n=2 (weak baryon decay scale, 6.3×10^{-10} s) and n=5 ($p\to e^+\pi^0$ partial lifetime scale, 2.6×10^{34} yr; current Super-K bound: $>2.4\times10^{34}$ yr at 90% CL). Black hole entropy receives a universal −ln 2 correction from the Z_2 seam. The Spinor Phase Gate $\Pi_Z(\varphi) = \sin^2(\varphi/2)$ is derived from j=1/2 uniqueness and SU(2) representation theory, eliminating the last free parameter in the Z-Sim forward simulator (ZS-T3 v1.0).

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python/mpmath, 50-digit precision) is publicly available.

Appendix

A.1 Multi-Layered Falsification Gates

Layer	Gate ID	Falsification Condition
Theoretical	F-M3.1	δφ = A falsified: Regge deficit ≠ A·I_cell for primitive cells.
Observational	F-M3.2	$p\to e^+\pi^0$ partial lifetime $\tau(p\to e^+\pi^0) < 10^{34}$ yr or $> 10^{35}$ yr. Current Super-K bound: $>2.4\times10^{34}$ yr (90% CL). Hyper-K target: $\sim10^{35}$ yr.
Observational	F-M3.3	T_local finite at Kerr horizon contradicts Z-anchor Tolman mechanism.

Modification	F-M3.4	Non-primitive cells dominate: I_cell ≠ 1 for the majority of cells.
Theoretical	F-M3.5	SU(2) Wigner d-matrix for j=1/2 contradicted (would invalidate all of QM).

References

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Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0.)